

Chapter 3

Critical Reflection Fundamentals: The Standards of Critical Thinking

What is Critical Thinking?

At the end of Chapter 2 we considered the uniqueness of learning through reflection on experience compared to other types of learning. One thing that is not unique about *critical* reflection more specifically, however, is that it uses the same standards that should characterize all good reasoning. These are the Standards of Critical Thinking.

As defined by Richard Paul, from the Foundation for Critical Thinking
(www.criticalthinking.org):

**"Critical Thinking is a systematic way to form and shape one's thinking....
It is thought that is disciplined, comprehensive, based on intellectual standards,
and as a result, is well-reasoned."**

This definition applies to any kind of thinking, including the kind that you'll be doing as you reflect critically on your service-learning experiences.

In fact, as you become more familiar with the Standards of Critical Thinking you will probably find that applying them improves a lot of what you do, including the thinking (and learning) that you do in all of your classes (and, of course, in life more generally).

Un-critical Thinking: An Example

Perhaps the best way to introduce the concept of critical thinking is with some examples of "un-critical" thinking. Let's start with a short one. The words in **bold** in each part of the critique that follows represent a specific critical thinking standard that we will discuss further later.

"Everybody knows that smoking is bad, so parents shouldn't smoke when they have children; those parents who do smoke obviously just don't care about their kids."

Critique:

"[S]moking is bad." What do you mean by "bad?" Bad for the children's health right now because they are inhaling second-hand smoke? Bad because the parents are modeling behavior that, if imitated by the children later on in life, could harm their children's health? Bad because the parents are modeling an unhealthy behavior that could send a message to their children that it's okay to engage in risky activities more generally?

- Failure to provide that **clarity** makes it hard to develop the learning much further. Clarity, therefore, can be an important "gate keeper" to learning.

"Everybody knows..." How do you know this? What is your evidence?

- A failure to provide evidence to support the **accuracy** of a claim can undermine the claim itself and any conclusions that are drawn from it. The student could have cited the Surgeon General's warnings that have been on cigarette packages for years, along with hundreds of public service announcements on TV, articles in the mass media, etc. While this is not likely to be a debatable statement (except to some plaintiffs' lawyers!) it still needs documentation so that its accuracy can be confirmed.
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"[T]hose parents who do smoke obviously just don't care about their kids." Is it really that simple? What are some of the complexities here? For example, nicotine is addictive so it can be very hard to quit. Smoking cessation medications, such as nicotine patches, are very expensive and the parents in question may not be able to afford it. Is it fair to single out this one behavior as evidence of a parent's attitudes? Is this the only or most important criterion against which to judge parental concern?

- Suggesting that all people who smoke around their children do not care about their children is the kind of oversimplification that should raise a red flag about the **depth** of someone's thinking.
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Clarity, **accuracy**, and **depth** are three of the critical thinking standards we will introduce you to in this chapter.

You will notice that each part of the critique begins with a set of questions:

- "What do you mean by...?" [**Clarity**]
- "How do you know this? What is your evidence?" [**Accuracy**]
- "Is it really that simple? What are some of the complexities here?" [**Depth**]

Let's look at a slightly longer example, and then we'll consider all of the standards, along with their associated questions, in greater detail.

Un-critical Thinking: Another Example

Being effective at improving the lives of a group of disadvantaged people (in this case the academic performance of underprivileged children) includes not only being good at the specific task or activity we are bringing to the community (explaining academic concepts to them), but also making them feel special and loved. This helps to make them more receptive to our efforts. It became clear that the more we got to know the kids and showed them that we actually cared about them, the more willing they were to pay attention to us and their homework. For example, when I first started working with William, he wouldn't even look at me or acknowledge my presence. But every week I brought him little gifts, like candy and stickers, and slowly he began to warm up to me so that now he runs up when I come in and is ready to get to work... This matters because underprivileged children are not getting the attention that they need at home....

Critique:

Note that while this example is stronger than the previous one, it still demonstrates superficial and problematic reasoning.

- The author of this excerpt does a better job of explaining what she means by some of the terms she uses (e.g., what she means by “disadvantaged people” and by the “task or activity we are bringing to the community”) and of offering examples to support her thinking (e.g., “For example, when I first started working with William ...” and “... little gifts, like candy and stickers, ...”). There is more **clarity** in this excerpt than in the previous one.

“[U]nderprivileged children are not getting the attention that they need at home.” What is the evidence for this claim? Did you actually see the children interacting with family members?

- This failure to provide evidence to support the **accuracy** of a claim allows an “elitist” point of view—that poor people don’t love their children as much as wealthier people—to go unchallenged.

Bringing presents means we care. What additional questions does this raise? What are the complexities here? Why is it that bringing gifts means that you care? Are there implications for equating gift-giving with caring, especially in an educational setting?

- That the student assumes that gift-giving is synonymous with a demonstration of caring without taking the time to question that assumption represents a lack of **depth** in her thinking.

Poor children lack attention and are therefore unmotivated to learn; bringing them gifts is a good way to show them that we care, and that motivates them to learn. Is there another way to look at this? Might any child (not just one from a disadvantaged background) be shy with strangers at first, and become more attentive and responsive over time as he/she gets to know the tutor?

- In this case, no alternative explanation is sought for the student's experience. Put yourself in William shoes: you are a 12 year-old boy with little interest in school who as a result has been made to attend this after school program, and one day a college-aged girl you've never met before shows up and suddenly wants to be your friend and tries to make you do your homework. Failing to think about the situation from someone else's perspective, especially someone directly involved in the experience, represents a lack of **breadth** in her thinking.

Why Does Critical Thinking Matter?

There are several fundamental problems with what the student in the previous example suggests that she learned.

First, she came to her conclusion that under-privileged children lack attention by assuming an association between levels of socio-economic status and levels of parental affection. That is, she assumed that poor children are loved less by their parents than the more well-off.

This is a good example of learning what might be the *wrong* thing from un-critical reflection on experience, not unlike Mark Twain's cat from the first chapter of the tutorial.

The learning from this student, however, is an even better example of a "problematic" learning. That's because through her role as a member of a larger community she has the ability to influence the lives of others, by both her actions and her words, either of which could be flawed if they are based on the conclusions she draws from her un-critical thinking.

Second, she left **un**-explored several potentially interesting concepts. In particular, she could have re-oriented her reflection around the nature of motivation, particularly as it relates to young children in an educational setting. Is giving gifts a form of bribery to encourage attention to schoolwork? How does that affect development of the intrinsic motivators necessary for long-term success? She might have drawn on her own experience as a student, or that of her peers, and even sought out the views of the tutorial staff and/or her college instructors.

This is a good example of what TS Eliot meant when he said that you might have had the experience, but *missed the meaning*.

Not Missing the Meaning

How can you make sure that your reflection on experience really leads to learning that is as meaningful and significant as possible? And just as important, how can you make sure that it doesn't produce problematic learning (as in the prior examples)?

What you need is a way to continuously check the quality of the thinking that is leading to your learning as you go along.

- The *Standards of Critical Thinking* will help you do just that.

What are the Standards of Critical Thinking?

Critical Thinking can be viewed as the application of the following standards:

Clarity, Accuracy, Precision, Relevance, Depth, Breath, Logic, Significance, and Fairness¹

¹Paul, R & Elder, L. 2001. *The Miniature Guide to Critical Thinking*. The Foundation for Critical Thinking. Santa Rosa, CA. www.criticalthinking.org

Read through the descriptions of each standard that follow. You will likely find that many of the questions are similar to the comments your instructors might write in the margins of your papers as they try to help you improve the quality of your thinking.

Clarity: Do you expand on ideas, express ideas in another way, provide examples or illustrations where appropriate?

- Lack of *clarity* is characterized by comments such as: For example? Which are...? What do you mean by that? Could you elaborate further?

Accuracy: Are all of your statements, is all of your information, factually correct and/or supported with evidence? Be careful not to make unsubstantiated claims. Make sure that you distinguish between what you think you know to be true even though you have no actual evidence, and what you can actually back up.

- Lack of *accuracy* is characterized by comments such as: How do you know this? Is this true? How could you check on that or verify it?

Precision: Do all of your statements or claims contain specific information, descriptions, or data?

- Lack of *precision* is characterized by comments such as: Could you give more details? Can you be more specific?

- Relevance:** Are all of your statements relevant to the question at hand? Does what you're saying connect to your central point?
- Lack of *relevance* is characterized by comments such as: How does this relate to the issue being discussed? How does this help us/you deal with the issue being discussed?
- There is a second component to relevance that, as you'll see, is specific to the structure of the reflective essays that you'll be writing: Are all of your statements relevant to the specific category of learning goal being discussed?
- Depth:** Do you explain the reasons behind your conclusions, anticipate and answer the questions that your reasoning raises and/or acknowledge the complexity of the issue?
- Lack of *depth* is characterized by comments such as: Why is that so? What are some of the complexities here? What would it take for that to happen? Would it be easy to do?
- Breadth:** Are you considering alternative points of view? Have you thought about how someone else might have interpreted the situation?
- Lack of *breadth* is characterized by comments such as: Would this look the same from the perspective of...? Is there another way to interpret what this means?
- Logic:** Does your line of reasoning make sense? Does it follow from the facts and/or what you said?
- Lack of *logic* is characterized by comments such as: Does what you said at the beginning fit with what you concluded at the end? Do your conclusions match the evidence that you have presented?
- Significance:** Do your conclusions or goals represent a (the) major issue raised by your reflection on experience?
- Lack of *significance* is characterized by comments such as: Is this the most important issue to focus on? Is this most significant problem to consider?
- Fairness:** Do you represent perspectives other than your own with integrity (without bias or distortion)?
- Lack of *fairness* is characterized by comments such as: Is this really what he/she believes? Would he/she agree with the way you have represented his/her point of view? Are you being true to this idea in your characterization of it?

The Standards of Critical Thinking Work Together

These standards do not work independently of each other.

As noted by Paul et al., **clarity** is the “gateway” standard. If a statement is unclear because it lacks examples or illustrations then it is hard to know if it is accurate or relevant. Consider the statement:

“The health of the American public is in jeopardy”

It’s hard to evaluate this statement because we don’t know what is meant by “health.” Health could be assessed by looking at life expectancy. But life expectancy is at an all-time high in the US and continues to increase, primarily because of improvements in health care. Perhaps the speaker is thinking about more specific measures of health, such as the incidence of obesity or diabetes, both of which have increased dramatically in recent years.

And if a statement isn’t **accurate**, or if supporting evidence isn’t **relevant**, then it is hard to know if the right questions have been asked or if the most significant learning is being addressed. In other words, the learning itself may be problematic, as we have already seen. Consider the statement:

“The majority of Americans eat at fast food restaurants at least once a week.”

What is the source of or evidence for this claim? If it isn’t accurate then any conclusions or learnings that are drawn from it, for example that fast food restaurants are a significant contributor to the obesity problem in this country, are called into question.

Even if you had the source, it would be helpful to know what is meant by “the majority.” Is that a simple majority, as in 51%, or a larger proportion, like 70-80%? Having that detail gives you greater **precision**, which in turn allows you to better assess the conclusions that are being drawn.

Keep in mind however that a statement can be clear, accurate and relevant but still not lead to a statement of learning that has **depth**. Consider the statement:

“If you want to lose weight all you have to do is eat less and exercise more.”

This statement ignores the **complexity** of body weight management, an issue that includes biological, psychological, and sociological components. If weight loss and maintenance of lost weight was so “easy,” more people who wanted to lose weight would be able to do so. So why aren’t they???

- Therefore it is always important to be checking your thinking using all of the standards.

One Additional Component

We are going to add one additional component to this list. Although it is not technically part of the Standards of Critical Thinking as developed by the Foundation for Critical Thinking, it is key to making sure that your SL-related learning is focused and well-articulated.

Integration: Do you make clear the connection between the service experience and the learning being articulated?

- Lack of *integration* is characterized by comments such as: How is this related to your service experience?

We are using the term integration here to refer to the critical connection or integration between the service-related experience and the learning being described. Because you are engaged in reflection on *service*, we want to make sure that you keep your thinking focused on what you have experienced in the context of your service-learning activities.

The table on the next page summarizes each of these standards and their associated questions. We encourage you to keep it handy as you reflect until you become familiar with them.

Critical Thinking Standards Table

Critical Thinking Standard	Description	Associated questions to ask to check your thinking
Integration	<i>Service experience clearly related to the learning</i>	Have I clearly shown the connection between my experience and my learning?
Clarity	<i>Expands on ideas, express ideas in another way, provides examples or illustrations where appropriate.</i>	- Did I give an example? - Is it clear what I mean by this? - Could I elaborate further?
Accuracy	<i>All statements are factually correct and/or supported with evidence.</i>	- How do I know this? - Is this true? - How could I check on this or verify it?
Precision	<i>Statements contain specific information</i>	- Can I be more specific? - Have I provided sufficient detail?
Relevance	<i>All statements are relevant to the question at hand; all statements connect to the central point.</i>	- How does this relate to the issue being discussed? - How does this help us/me deal with the issue being discussed?
Depth	<i>Explains the reasons behind conclusions and anticipates and answers the questions that the reasoning raises and/or acknowledges the complexity of the issue.</i>	- Why is this so? - What are some of the complexities here? - What would it take for this to happen? - Would this be easy to do?
Breadth	<i>Considers alternative points of view or how someone else might have interpreted the situation.</i>	- Would this look the same from the perspective of...? - Is there another way to interpret what this means?
Logic	<i>The line of reasoning makes sense and follows from the facts and/or what has been said.</i>	- Does what I said at the beginning fit with what I concluded at the end? - Do my conclusions match the evidence that I have presented?
Significance	<i>The conclusions or goals represent a (the) major issue raised by the reflection on experience.</i>	- Is this the most important issue to focus on? - Is this most significant problem to consider?
Fairness	<i>Other points of view are represented with integrity (without bias or distortion)</i>	Have I represented this viewpoint in such a way that the person who holds it would agree with my characterization?

Modified source: Paul, R & Elder, L. 2001. The Miniature Guide to Critical Thinking. The Foundation for Critical Thinking. Santa Rosa, CA. www.criticalthinking.org

One Last Example...

We are going to wrap up this discussion with one final example of un-critical thinking that we heard a colleague share at a service-learning conference:

A college student was describing how wonderful and personally transforming his service-learning experience had been—one that involved working with a disadvantaged population. At the end of his reflective essay he said that he very much hoped that some day his own children "could have a similar opportunity."

Think for a minute about the implications of what he said. For this "wish" to come true, there would have to be a similarly disadvantaged population in the future with whom his children could work. Might the better "wish" be that by the time his children were in their 20s society had changed such that there would not be a need for such an activity, or at least less of a need?

- Because the student didn't think about the *implications* of what he said (a problem of **depth**) he produced a learning only from the perspective of what *he* got out of the experience, rather than thinking about its more complex civic dimensions.
- We can practice **breadth** here and note that the student may have meant that he hoped his children could have a similarly valuable learning opportunity, not that he hoped they would be able to work with disadvantaged people; note how the lack of **clarity** in the phrase "could have a similar opportunity" can easily lead to multiple, perhaps unintended, interpretations.

We hope that this tutorial will support you in avoiding similar shortcomings in reasoning, helping you to not miss the meaning.

Critical Thinking Now and in the Future

Critical thinking will be key to getting all that you can out of your service-learning experience.

But critical thinking can be key in making sure that you get the most out of ALL other aspects of your life as well – academically, professionally, and personally.

We think you'll find that considering these questions will be helpful in *everything* you do — now and into the future — from writing a paper or answering an essay question for another class, to writing a cover letter for a job or a personal statement for graduate school, to making decisions about whom to vote for or even how to raise your children!

We will give you just one example on the next page. We think that you will find many more examples as you come to "internalize" the standards themselves.

Learning to think critically beyond service-learning: An example

Read the excerpt from a medical school essay below. See if you can apply some of the Standards of Critical Thinking to improve its quality.

Tribulation is not selective. At some point in our lives we are all faced with obstacles to overcome. Everyone is susceptible to disease or illness and doctors offer assistance to those in need. As a physician I will be able to play a central role in making decisions that directly affect patient care and treatment. In order to do this a doctor must attain the knowledge necessary to be effective.

I work in a diabetes clinic where we visit local communities once a week that have a shortage of medical care. In these rural areas we do not just treat diabetes, but people with other major health problems as well. Having this experience has made me appreciate a more generalist approach to medicine and therefore feel that I will be best suited to being a general practitioner. My additional training in public health will help me help my patients make contributions to their own well-being. Ideally, I would be involved in both the traditional primary care of patients and the preventative healthcare awareness of the community.

Recent issues surrounding preventative health are motivating to me. I think that it is very important for physicians, especially in this region, to be aware of the indicators of chronic illness. As a physician I hope to educate and employ treatment plans in order to proactively combat illness and maintain the health of my patients. It is a primary care physician's responsibility to facilitate healthcare learning in the community as they are the principal source of knowledge. The X School of Medicine offers the type of community focused education that will best prepare me for this career path.

Now re-read the essay with the Critical Thinking questions embedded. You can see how much stronger it would be if the writer had been keeping these in mind as she composed it. In particular, her essay would move from one of superficiality that would likely earn a quick rejection from an admissions committee to a more thoughtful discourse on both the opportunities that she looks forward to and the challenges she knows she will face in her chosen profession.

Tribulation is not selective. At some point in our lives we are all faced with obstacles to overcome. Everyone is susceptible to disease or illness and doctors offer assistance to those in need. As a physician I will be able to play a central role in making decisions that directly affect patient care and treatment. In order to do this a doctor must attain the knowledge necessary to be effective. Do the first sentences tie clearly to the last? (Logic)

I work in a diabetes clinic where we visit local communities once a week that have a shortage of medical care. In these rural areas we do not just treat diabetes, but people with other major health problems as well. For example? (Clarity). Having this experience has made me appreciate a more generalist approach to medicine Why is this so? (Depth) and therefore feel that I will be best suited to being a general practitioner. My additional training in public health will help me help my patients make contributions to their own well-being. Can you be more precise? (Precision) Ideally, I would be involved in both the traditional primary care of patients and the preventative healthcare awareness of the community. What specifically would you be doing? (Precision)

Recent issues surrounding preventative health are motivating to me. For example? (Clarity) I think that it is very important for physicians, especially in this region, to be aware of the indicators of chronic illness. Why is this so? (Depth) As a physician I hope to educate and employ treatment plans in order to proactively combat illness and maintain the health of my patients. It is a primary care physician's responsibility to facilitate healthcare learning in the community as they are the principal source of knowledge. Will this be easy to do? What are the complexities? (Depth)

Chapter 3 Summary

In brief, here is what we've learned from Chapter 3 --

The *Standards of Critical Thinking* are key to helping you have the experience but not miss the meaning by allowing you to continuously check the quality of the thinking that is leading to your learning. You will find them to be helpful now as well as throughout your life.

The standards are:

- Integration:** Are all of your statements relevant to the specific category of learning goal being discussed?
- Clarity:** Do you expand on ideas, express ideas in another way, provide examples or illustrations where appropriate?
➤ For example? Which are...? What do you mean by that? Could you elaborate further?
- Accuracy:** Are all of your statements, is all of your information, factually correct and/or supported with evidence? Be careful not to make unsubstantiated claims. Make sure that you distinguish between what you think you know to be true even though you have no actual evidence, and what you can actually back up.
➤ How do you know this? Is this true? How could you check on that or verify it?
- Precision:** Do all of your statements or claims contain specific information, descriptions, or data?
➤ Could you give more details? Can you be more specific?
- Relevance:** Are all of your statements relevant to the question at hand? Does what you're saying connect to your central point?
➤ How does this relate to the issue being discussed? How does this help us/you deal with the issue being discussed?
- Depth:** Do you explain the reasons behind your conclusions, anticipate and answer the questions that your reasoning raises and/or acknowledge the complexity of the issue?
➤ Why is that so? What are some of the complexities here? What would it take for that to happen? Would it be easy to do?
- Breadth:** Are you considering alternative points of view? Have you thought about how someone else might have interpreted the situation?
➤ Would this look the same from the perspective of...? Is there another way to interpret what this means?
- Logic:** Does your line of reasoning make sense? Does it follow from the facts and/or what you said?
➤ Does what you said at the beginning fit with what you concluded at the end? Do your conclusions match the evidence that you have presented?
- Significance:** Do your conclusions or goals represent a (the) major issue raised by your reflection on experience?
➤ Is this the most important issue to focus on? Is this most significant problem to consider?
- Fairness:** Do you represent perspectives other than your own with integrity (without bias or distortion)?
➤ Is this really what he/she believes? Would he/she agree with the way you have represented his/her point of view? Are you being true to this idea in your characterization of it?

Check Your Understanding / Practice What You've Learned

Now that you're familiar with the Standards of Critical Thinking, you can begin to practice applying them. Read the example below and our evaluation of it in light of the standards. Then similarly evaluate the rest of the quotes using the Standards of Critical Thinking table. Make sure that you identify which question (or even questions) you think are in need of being answered.

- Note that for many, if not most of these examples, there could be a number of "right" answers. That's because as we've seen, the standards are interconnected. So your focus should not be on whether you "picked the right standard" but rather on becoming familiar with the types of questions you should be asking in order to check the quality of your thinking.

Example

"Each member of our group shares a common interest in that we all like kids, therefore we should be successful in our efforts because we are all eager to work with children."

Critique

"...because..." Does this line of reasoning make sense? The writer has concluded that the project should be successful based only on the group's mutual interest in children, rather than considering the wide variety of skills (personal and interpersonal), knowledge, etc., that are important as well. This is faulty reasoning, or a problem of **logic**.

1. *If the government was involved somehow, then the program would be more structured.*
2. *I came to realize that racism is learned by talking to the younger children versus the older children. The younger ones were oblivious to my skin color. On the other hand, the older children realized that my skin color was different and were therefore less inclined to talk to me.*
3. *I was able to connect with these kids and come to care about them. I now understand how they adapt to their surroundings, how they communicate with each other, and how they can work together when needed to solve problems as a group. If this can happen with a small group of kids then this can happen with adults who may not agree or come from the same backgrounds.*
4. *I was not used to working with so many kids. This made me nervous and I do not like being nervous. This is important to know because I know that in the future I do not want to have a job that requires me to work with children. After I worked with them for a few weeks and got to know them the nervousness wore off and I became comfortable. This was important because it allowed me to be the best role model I could be.*
5. *I realize now that I can incorporate fun into any task.*